AI BUSINESS ANALYSIS PFC PROGRESS

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System Objectives

- Provide a secure login system for users (business owners) We added this feature for proper security.
- Visualize business performance through dashboards
- Store and manage business data efficiently using a relational database
- Perform basic data analysis to generate insights (sales trends, product performance)
- Separate frontend and backend logic for scalability and maintainability

Current Progress

Frontend Development

- Completed `dashboard.html`
- Completed `style.css`
- Built initial UI structure including:
  - Navigation sidebar
  - Dashboard layout
  - Metrics display sections
  - Placeholder areas for charts and analytics

Backend Planning (PHP)

- Backend architecture defined using PHP
- Planned features:
  - User authentication (login/register system)
  - Session-based access control
  - Secure data handling between frontend and database
- Responsibility division:

    - PHP → backend logic and database interaction
    - HTML/CSS/JS → frontend interface and interactivity

Database Design (ERD) & UML

An Entity Relationship Diagram (ERD) was designed to structure the database for the system.

Main Entities Identified:

1. Users

- user_id (PK)
- name
- email
- password
- role (admin/business owner)
- created_at

2. Products

- product_id (PK)
- product_name
- category
- price
- stock_quantity
- user_id (FK)

3. Sales

- sale_id (PK)
- product_id (FK)
- quantity_sold
- sale_date
- total_amount

Relationships:

- One user can have multiple products
- One product can have multiple sales records
- Sales is dependent on Products (foreign key relationship)

This structure supports tracking product performance and generating analytics per user/business.

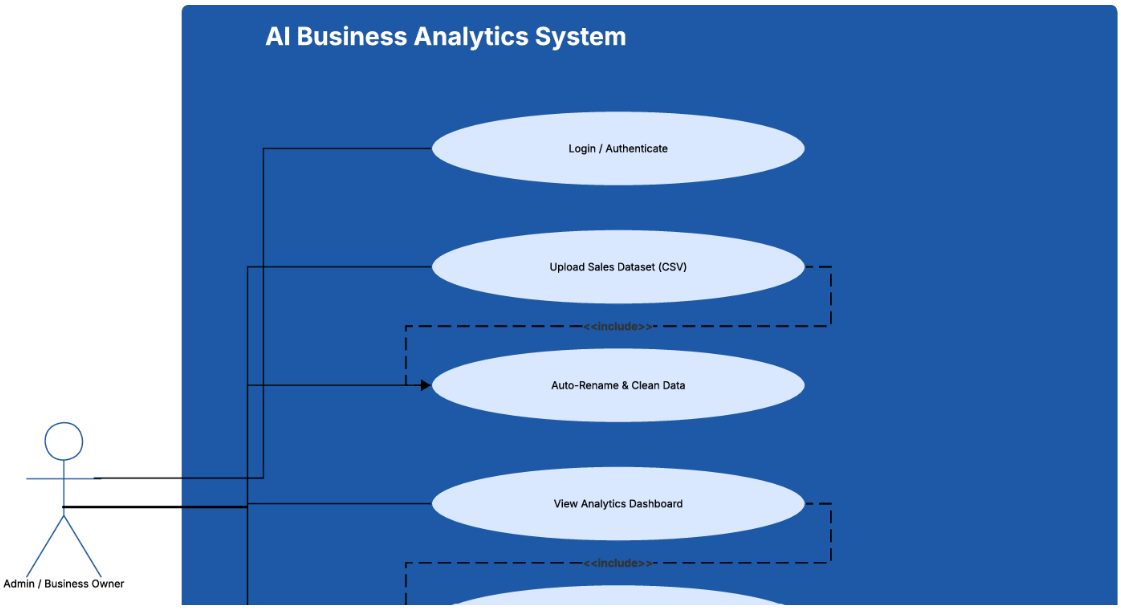


Figure 1: use case diagram

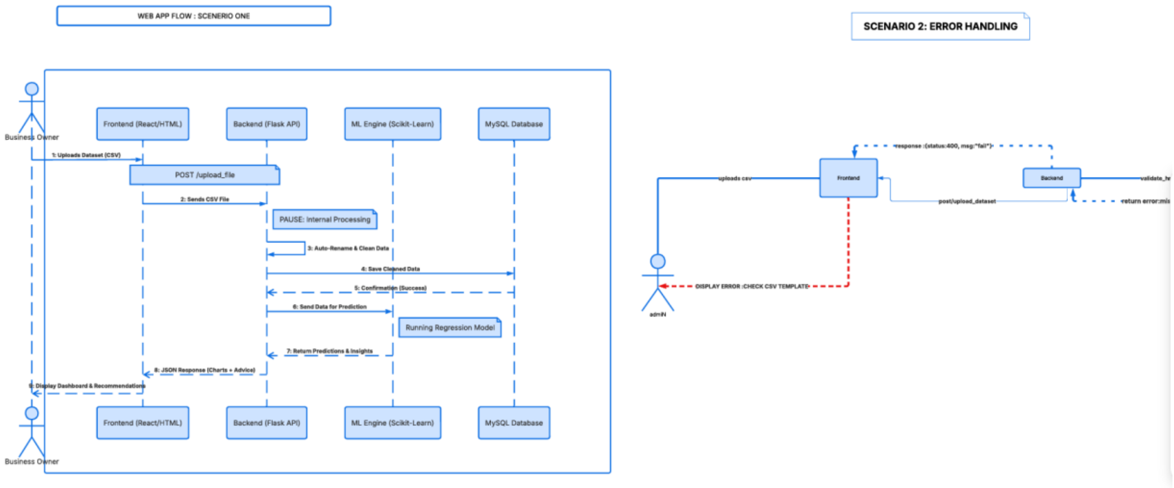


Figure 2: sequence diagram

Data Analysis Work Completed

We performed initial analysis on business/product data to guide dashboard design.

Key Insights Derived:

- **Product performance tracking**
  - Identified best-selling products based on sales volume
- **Revenue estimation**
  - Calculated total revenue using: quantity sold × price
- **Category comparison**
  - Compared performance across different product categories
- **Trend observation**

- Identified patterns in sales over time (basic time-based grouping)

Dashboard Metrics Derived from Analysis:

- Total Sales Revenue
- Top Performing Product
- Total Products Sold
- Category Performance Breakdown

Dashboard Development

The dashboard was designed based on the analysis results.

Features Implemented (UI Level):

- Summary cards for key metrics
- Layout for charts (future integration with dynamic data)
- Sections prepared for:
  - Sales overview
  - Product performance
  - Category insights

Design Approach we will implement:

- Simple, clean, and data-focused interface
- Modular layout to allow future integration with PHP and database
- Responsive structure (basic CSS framework implemented)

System Architecture (Planned Final Structure)

- **Frontend:** HTML, CSS, JavaScript
  → Handles UI and dashboard interactions
- **Backend:** PHP
  → Handles authentication, logic, and data processing
- **Database:** MySQL
  → Stores users, products, and sales data
- **Analytics Layer:**
  → Queries + calculations for dashboard insights

Work Completed So Far

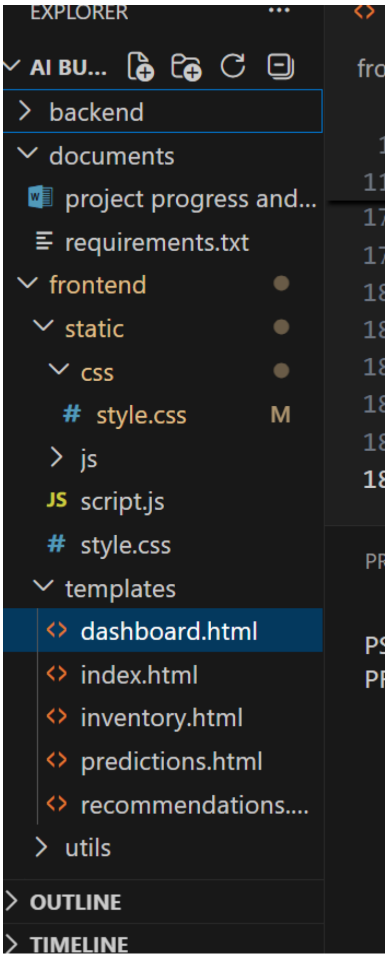
- Dashboard UI completed (HTML + CSS)
- Initial ERD designed
- Basic data analysis performed
- System architecture defined
- Backend responsibilities assigned (PHP structure planned)

- GitHub repository set up for collaborative development using push/pull workflow

Next Steps

- Implement PHP backend:
  - User registration/login system
  - Session management
- Create MySQL database based on ERD
- Connect frontend dashboard to backend data
- Replace static dashboard values with dynamic analytics
- Add JavaScript interactivity (charts, filters, updates)
- Improve data visualization (graphs/charts integration)
- Update the dashboard static page to a dynamic page

- ## Folders organization



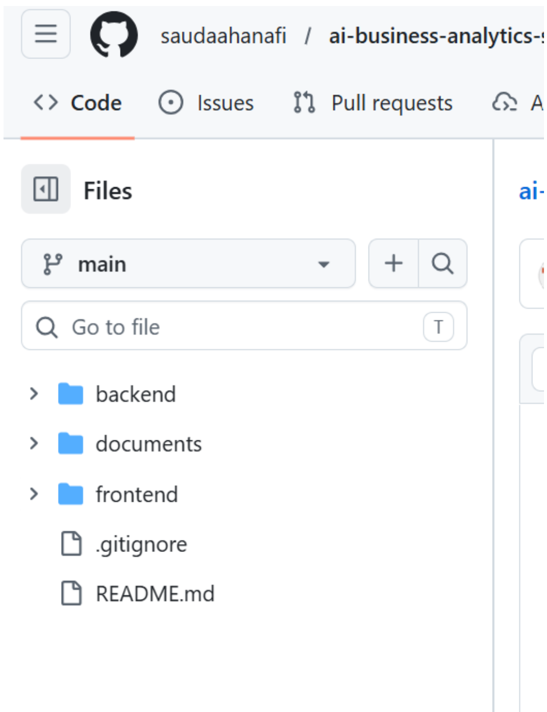


Figure 3: file organization

we have these files organized in the github repository for an organized workflow between the members working on frontend and backend simultaneously.

10. Challenges & Considerations

- Ensuring secure authentication and session handling
- Transitioning from static dashboard to dynamic data-driven system
- Maintaining clean separation between frontend and backend
- Ensuring database design supports scalable analytics

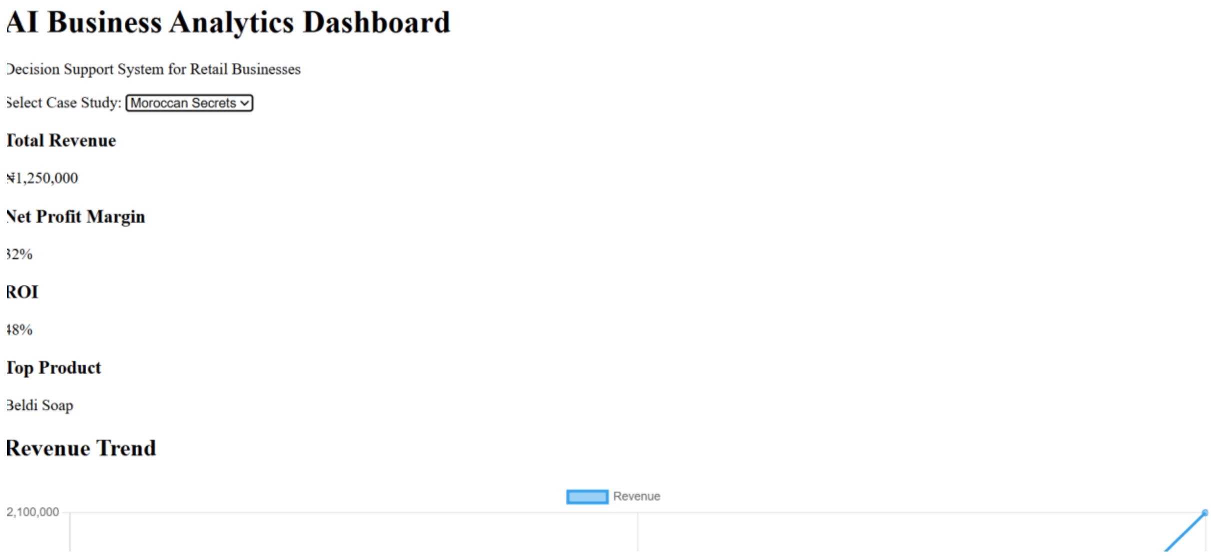


Figure 4: static dashboard page

11. Summary

The project is progressing well with frontend development completed and backend design in progress. The ERD and data analysis phase has helped define the system structure and dashboard metrics. The next phase focuses on implementing PHP backend functionality and integrating the database to enable dynamic analytics and ensure data security.